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Brake system with normal non-failure and manual front push-through modes

Abstract

A brake system includes a reservoir and a master cylinder operable to provide a brake signal responsive to actuation of a brake pedal connected thereto. A power transmission unit is configured for selectively providing pressurized hydraulic fluid for actuating at least one of the pair of front wheel brakes and the pair of rear wheel brakes in a normal non-failure braking mode. A iso/dump valve arrangement is hydraulically interposed between the power transmission unit and a corresponding one of the wheel brakes. A fluid separator is hydraulically interposed between a selected one of the pair of front wheel brakes and a corresponding iso/dump valve arrangement. A normally open base brake iso valve is energized closed during normal non-failure braking mode. The base brake iso valve is open to place the master cylinder into fluid communication with at least one of the fluid separators during manual push-through braking mode.

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Background/Summary

TECHNICAL FIELD

(1) This disclosure relates to an apparatus and method for use of a brake system and, more particularly, to a method and apparatus of a hydraulic brake system for actuating at least one of a pair of front wheel brakes and a pair of rear wheel brakes in normal non-failure and manual push-through modes.

BACKGROUND

(2) A brake system may include anti-lock control including a pedal-operated hydraulic braking pressure generator, a braking pressure modulator which is provided in the pressure fluid conduits between the braking pressure generator and the wheel brakes and which serves to vary the braking pressure by changing the volume of a chamber containing the hydraulic fluid, sensors for determining the wheel rotational behavior, and electronic circuits for processing the sensor signals and for generating braking-pressure control signals. Brake systems may also include both anti-lock control and traction slip control, which can use braking pressure modulators for controlled vehicular braking.

(3) Descriptions of prior art brake systems are in U.S. Pat. No. 10,730,501, issued 4 Aug. 2020 to Blaise Ganzel and titled “Vehicle Brake System with Auxiliary Pressure Source”, and in U.S. Patent Application Publication No. 2020/0307538, published 1 Oct. 2020 by Blaise Ganzel and titled “Brake System with Multiple Pressure Sources”, both of which are incorporated herein by reference in their entirety for all purposes.

(4) It may be desirable for a brake system to include various features to facilitate continued operation of at least one of the wheel brakes even after a partial or complete failure of another portion of the brake system.

SUMMARY

(5) In an aspect, a brake system for actuating at least one of a pair of front wheel brakes and a pair of rear wheel brakes in normal non-failure and manual push-through modes is disclosed. The system includes a reservoir and a master cylinder operable to provide a brake signal responsive to actuation of a brake pedal connected thereto, the master cylinder being selectively operable during a manual push-through mode by actuation of the brake pedal to generate brake actuating pressure to at least one output for hydraulically actuating the pair of front wheel brakes. A power transmission unit is configured for selectively providing pressurized hydraulic fluid for actuating at least one of the pair of front wheel brakes and the pair of rear wheel brakes in a normal non-failure braking mode. A plurality of iso/dump valve arrangements is provided. Each iso/dump valve arrangement is hydraulically interposed between the power transmission unit and a corresponding one of the wheel brakes. A plurality of fluid separators is provided. Each fluid separator is hydraulically interposed between a selected one of the pair of front wheel brakes and a corresponding iso/dump valve arrangement. A normally open base brake iso valve is provided. The base brake iso valve is energized closed during normal non-failure braking mode. The base brake iso valve is open to place the master cylinder into fluid communication with at least one of the fluid separators during manual push-through braking mode. An electronic control unit is provided for selectively controlling at least one of the power transmission unit, at least one of the plurality of iso/dump valve arrangements, and the base brake iso valve responsive to the brake signal.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) For a better understanding, reference may be made to the accompanying drawings, in which:

(2) FIG. 1 is a schematic hydraulic diagram of a brake system according to an aspect of the present invention, in a first configuration; and

(3) FIG. 2 is a schematic hydraulic diagram of the brake system of FIG. 1, in a second configuration.

(4) FIG. 3 is a schematic hydraulic diagram of a brake system according to an aspect of the present invention, in a first configuration; and

(5) FIG. 4 is a schematic hydraulic diagram of the brake system of FIG. 3, in a second configuration.

DESCRIPTION OF ASPECTS OF THE DISCLOSURE

(6) Unless defined otherwise, all technical and scientific terms used herein have the same meaning as is commonly understood by one of ordinary skill in the art to which the present disclosure pertains.

(7) The invention comprises, consists of, or consists essentially of the following features, in any combination.

(8) FIGS. 1-2 schematically depict a brake system **100** for actuating a pair of front wheel brakes and a pair of rear wheel brakes, in first and second configurations, respectively. The brake system **100** is shown here as a hydraulic braking system in which fluid pressure is utilized to apply braking

forces for the brake system **100**. The brake system **100** may suitably be used on a ground vehicle, such as an automotive vehicle having four wheels with a wheel brake associated with each wheel. Furthermore, the brake system **100** can be provided with other braking functions such as anti-lock braking (ABS) and other slip control features to effectively brake the vehicle. Components of the brake system **100** may be housed in one or more blocks or housings. The block or housing may be made from solid material, such as aluminum, that has been drilled, machined, or otherwise formed to house the various components. Fluid conduits may also be formed in the block or housing.

(9) In the illustrated embodiment of the brake system **100**, there are four wheel brakes **102A**, **102B**, **102C**, and **102D**. The wheel brakes **102A**, **102B**, **102C**, and **102D** can have any suitable wheel brake structure operated electrically and/or by the application of pressurized brake fluid. Each of the wheel brakes **102A**, **102B**, **102C**, and **102D** may include, for example, a brake caliper mounted on the vehicle to engage a frictional element (such as a brake disc) that rotates with a vehicle wheel to effect braking of the associated vehicle wheel. The wheel brakes **102A**, **102B**, **102C**, and **102D** can be associated with any combination of front and rear wheels of the vehicle in which the brake system **100** is installed. For example, the brake system **100** may be configured as shown, such that a power transmission unit **104** is configured for selectively providing pressurized hydraulic fluid for actuating at least one of the pair of front wheel brakes **102B** and **102D** and to the pair of rear wheel brakes **102A** and **102C**, in a non-failure normal braking mode, as will be discussed below. In the depicted embodiments of the brake system **100**, the power transmission unit **104** provides fluid to both of the pair of front wheel brakes **102B** and **102D** and the pair of rear wheel brakes **102A** and **102C**, in the normal non-failure braking mode. The brake system **100** is configured for actuating at least one of the pair of front wheel brakes **102B** and **102D** and the pair of rear wheel brakes **102A** and **102C** in both normal non-failure and manual push-through modes, as will be described herein.

(10) In this example, the wheel brake **102A** may be associated with a right rear wheel of the vehicle in which the brake system **100** is installed, and the wheel brake **102B** may be associated with the right front wheel. The wheel brake **102C** may be associated with the left rear wheel, and the wheel brake **102D** may be associated with the left front wheel. Alternatively, though not depicted here, the brake system **100** may be configured such that the wheel brakes **102A** and **102B** are associated with wheels at the front or rear axle of the vehicle, and the wheel brakes **102C** and **102D** are associated with wheels at the other axle of the vehicle. (It is contemplated that various other components, such as electric brake motors, could be reconfigured by one of ordinary skill in the art to accommodate such an altered system in certain use environments.)

(11) A reservoir **106** stores and holds hydraulic fluid for the brake system **100**. The fluid within the reservoir **106** is preferably held at or about atmospheric pressure, but the fluid may be stored at other pressures if desired. The reservoir **106** is shown schematically having two tanks or sections with fluid conduit lines connected thereto. The sections can be separated by several interior walls within the reservoir **106** and are provided to prevent complete drainage of the reservoir **106** in case one of the sections is depleted due to a leakage via one of the lines connected to the reservoir **106**. Alternatively, the reservoir **106** may include multiple separate housings. The reservoir **106** may include at least one fluid level sensor **108** for detecting the fluid level of one or more of the sections of the reservoir **106**.

(12) As shown schematically in FIG. 1, the brake system **100** includes a master cylinder **110** with a housing **112** defining a longitudinally extending bore for slidably receiving various cylindrical pistons and other components therein. Examples of such components are the MC springs (three shown) longitudinally extending at least partially in series within the bore, as shown in the Figures. Note that the housing **112** is not specifically schematically shown in the Figures, but instead the walls of the longitudinally extending bore are schematically illustrated. The housing **112** may be formed as a single unit or include two or more separately formed portions coupled together. For some use environments, the master cylinder **110** may be of a single circuit master cylinder type.

(13) A brake pedal **114** is operatively connected to the master cylinder **110** and is actuated by the

driver of the vehicle as the driver presses on the brake pedal **114**. A brake travel sensor **116** (two shown, for redundancy) is configured provide a brake signal to other portions of the brake system **100**, indicating a depression of the brake pedal **114** (binary on/off, and/or including some quantitative information related to the brake pedal **114** depression). That is, the master cylinder **110** is operable to provide a brake signal responsive to actuation of the brake pedal **114** connected thereto. The brake signal may be utilized by one or more other components of the brake system **100** to effectuate desired braking of the motor vehicle, such as via transmission of electronic signals when the brake system **100** is in a normal non-failure braking mode.

(14) The brake pedal **114** and related structures of the master cylinder **110** may also instead be used as a back-up source of pressurized fluid to essentially replace the normally supplied source of pressurized fluid from the power transmission unit **104** under certain failed conditions of the brake system **100**, and/or upon initial startup of the brake system **100**. This situation is referred to as a manual push-through event, or a “manual apply” and may be accomplished in coordination with actuation of any backup source of pressurized fluid available, or independently thereof. The master cylinder **110** is selectively operable during a manual push-through mode by actuation of the brake pedal **114** to generate brake actuating pressure to at least one MC output **118** for hydraulically actuating at least one of the wheel brakes **102** of the brake system **100**

(15) In such a manual push-through mode, the master cylinder **110** can supply pressurized fluid to the MC output **118**, which is then routed to the hydraulically operated pair of front wheel brakes **102B**, **102D** as desired. This flow is pushed through, largely under mechanical pressure upon the brake pedal **114** from the driver's foot, from the master cylinder **110**. That is, the master cylinder **110** is selectively operable during a manual push-through mode by actuation of the brake pedal **114** connected to the master cylinder **110** to generate brake actuating pressure for hydraulically actuating at least one of the pair of front wheel brakes **102B**, **102D** and the pair of rear wheel brakes **102A**, **102C**. (The below description presumes that the pair front wheel brakes **102B**, **102D** are the hydraulically actuated ones of the brakes in the manual push-through mode, for ease of description.)

(16) An MC primary piston **120** is connected with a brake pedal **114** via a linkage arm **122**. The MC primary piston **120** is slidably disposed in the bore of the housing **112** of the master cylinder **110**. When the brake pedal **114** is in its rest position (the driver is not depressing the brake pedal **114**), the structures of the master cylinder **110** permit at least a portion of fluid communication between the bore of the housing **112** and the reservoir **106** via a reservoir conduit **126**. Leftward movement (in the orientation of the Figures) of the MC primary piston **120** due to pressure by the driver's foot upon the brake pedal **114** may cause, under certain conditions, a pressure increase within the master cylinder **110**, which results in egress of pressurized fluid from the longitudinal bore of the master cylinder **110** via MC output **118**.

(17) A pedal simulator **124** may be in selective fluid communication with the master cylinder **110**. As shown, the pedal simulator **110** is integrated bodily into the master cylinder **110** to provide a desired level of fluid communication, but it is contemplated that the “selective fluid communication” could also or instead be provided via connection of a separately located pedal simulator (not shown) to the master cylinder **120** via one or more hydraulic passages. Regardless of the exact nature of connection, the pedal simulator **124** may be in selective fluid communication with the master cylinder **110** for providing predetermined brake pedal **114** response to the driver (e.g., brake pedal “feel”). For example, and as shown in the Figures, the pedal simulator **124** is integrated into the master cylinder **110**, and extends substantially coaxially with the longitudinally extending bore of the master cylinder **110**.

(18) The pedal simulator **124** shown in the Figures as being integrated into the master cylinder **110** includes a simulator piston **128** having at least one simulator port **130**. The simulator piston **128** travels reciprocally longitudinally within the housing **112** of the master cylinder **110** responsive to a driver's foot pressure upon the brake pedal **114** and/or fluid pressures and other portions of the

brake system **100**. When the simulator port **130** travels far enough leftward (in the orientation of the Figures) to be sealed off from fluid communication with the reservoir conduit **126**, the simulator piston **128** will normally substantially cease leftward travel and instead provide increased resistance to leftward travel of the brake pedal **114** and MC primary piston **120**. This provides desired fluid response and pedal simulator function within the master cylinder **110** itself.

(19) The housing **112** and other components associated with the master cylinder **110** may include any desired number and arrangement of seals, orifices, filters, passages, springs, resilient cushions (e.g., the shaded area of the Figures within the pedal simulator **124**), rods, spring cages, ports, and other structures for helping to achieve desired push-through capabilities and characteristics, pedal response “feel” for a driver, and pressurization/travel of hydraulic fluid within the brake system **100** as desired.

(20) The power transmission unit **104** is configured for selectively providing pressurized hydraulic fluid for actuating at least one of the pair of front wheel brakes **102B**, **102D** and the pair of rear wheel brakes **102A**, **102C** in a normal non-failure braking mode. That is, the power transmission unit **104** of the brake system **100** functions as a source of pressure to provide a desired pressure level to the hydraulically operated one(s) of the wheel brakes **102A**, **102B**, **102C**, and **102D** during a typical or normal brake apply. After a brake apply, fluid from the hydraulically operated ones of the wheel brakes **102A**, **102B**, **102C**, and **102D** may be returned to the power transmission unit **104** and/or be diverted to the reservoir **106**.

(21) In the depicted embodiments, the power transmission unit **108** is shown as being a dual acting plunger (“DAP”) type power transmission unit, though a single acting plunger (“SAP”) or any other desired type of controllable hydraulic fluid pressurizer could be also or instead provided to the brake system **100**. It is also contemplated that other configurations (not shown) of the brake system **100** could include hydraulic control of just a selected one of the pair of front wheel brakes and the pair of rear wheel brakes (with the other pair being electrically controlled/actuated). One of ordinary skill in the art would be readily able to provide such an arrangement for a desired use environment, following aspects of the present invention.

(22) The Figures also depict a replenishing check valve **132**, which is located fluidically between the reservoir **106** and the power transmission unit **104**. When present, the replenishing check valve **132** may be provided to assist with refilling of the power transmission unit **104** (or components thereof) under predetermined conditions. For example, the replenishing check valve **132** may help to facilitate refilling of the chamber in front of the DAP head when a DAP-type power transmission unit **104** is building pressure during its retraction stroke by pushing fluid out of the annular chamber behind the DAP head. This may be done, for example, during slip control if additional flow to the brakes is needed after the DAP is stroked fully forward.

(23) The brake system **100** also includes a plurality of iso/dump valve arrangements **134**. Each iso/dump valve arrangement **134** is hydraulically interposed between the power transmission unit **104** and a corresponding one of the wheel brakes **102**. The iso/dump valve arrangements **134** each include respective serially arranged iso and dump valves **136** and **138**. The iso valve **136** for each iso/dump valve arrangement **134** is located hydraulically between a respective wheel brake **102** and the power transmission unit **104**, and the dump valve **138** for each iso/dump valve arrangement **134** is located hydraulically between a respective wheel brake **102** and the reservoir **106**, for the corresponding wheel brake **102**. Here, for clarity in the Figures, the iso and dump valves **136** and **138** are appended with the letter “B” or “D”, referencing a respective front wheel brake **102B** and **102D**, or the letter “A” or “C”, referencing a respective rear wheel brake **102A** and **102C**, with which the so-labeled valves are respectively associated.

(24) Thus, the iso/dump valve arrangements **134** downstream of the power transmission unit **104** each include iso and dump valves **136** and **138** which are controlled between their open and closed positions to provide different braking pressures among the wheel brakes **102**, as desired for a particular braking situation. The iso/dump valve arrangements **134** may selectively provide slip

control to at least one wheel brake **102** of the pair of front wheel brakes **102B**, **102C** and the pair of rear wheel brakes **102A**, **102D** powered by the power transmission unit **104**. More broadly, the iso/dump valve arrangements **134**, and/or other valves of the brake system **100**, any of which may be solenoid-operated and have any suitable configurations, can be used to help provide controlled braking operations, such as, but not limited to, ABS, traction control, vehicle stability control, dynamic rear proportioning, regenerative braking blending, and autonomous braking, particularly during normal non-failure mode operation of the brake system **100**.

(25) As depicted in the Figures, the brake systems **100** shown therein may have at least one iso/dump valve arrangement **134** including a normally open dump valve **138A**, **138C** for each wheel brake **102A**, **102C** of the pair of rear wheel brakes. The brake systems **100** shown in the Figures may instead be configured with at least one iso/dump valve arrangement **134** including a normally closed dump valve **138B**, **138D** for each wheel brake **102B**, **102D** of the pair of front wheel brakes, as desired for a particular use environment. The normally open ones of the dump valves **138** may help to provide any desired vent path from the corresponding brakes **102** and/or the power transmission unit **104** to the reservoir **106**.

(26) A normally closed DAP valve **140** may be interposed hydraulically between the power transmission unit **104** and the plurality of iso/dump valve arrangements **134**. When present, the normally closed DAP valve **140** (which is usually energized open) may assist the power transmission unit **104** in acting like a single acting plunger type power transmission unit when the DAP valve **140** is open. The DAP valve **140** can be closed to invoke the “pumping action” rearward travel of the dual-acting plunger type power transmission unit **104**, in a known manner. If the normally closed DAP valve **140** fails—in the circuit shown in FIG. 1—the rear wheel brakes **102A**, **102C** may become unavailable under certain use conditions, but the driver will still be able to utilize the front brakes **102B**, **102D** in the manual push-through mode, via the master cylinder **110**.

(27) At least one fluid separator (two shown, at **142**) may be provided to the brake system **100**. When present, each fluid separator **142** is hydraulically interposed between a selected one of the pair of front wheel brakes **102B**, **102D** and a corresponding iso/dump valve arrangement **134B**, **134D**, as indicated by the “B” and “D” suffixes on the fluid separator **142B**, **142D**. Each fluid separator **142B**, **142D** may include a fluid separator piston **144** which is spring-biased toward a first end (the right end, in the orientation of the Figures) of a fluid separator chamber **146**. The fluid separator piston **144** travels reciprocally longitudinally between the first end and a second end (the left end, in the orientation of the Figures) of the fluid separator chamber **146** when the brake system **100** is in a normal non-failure braking mode, responsive to fluid pressure and other changes within the iso/dump valve arrangements **134** or other portions of the brake system **100**, occasioned by the power transmission unit **104**.

(28) In contrast, the fluid separator piston **144** is maintained substantially adjacent the first end of the fluid separator chamber **146** in manual push-through braking mode, such that a separator port **148** of each fluid separator piston **144** is available to place the separator conduit **150** in fluid communication with each of the pair of front wheel brakes **102B**, **102D**. Since the separator conduit **150** is in fluid communication with the master cylinder output **118**, the master cylinder **110**, as a result, may function as a source of pressure to provide a desired pressure level in a manual push-through mode of operation to at least a selected one of the pair of front wheel brakes **102B** and **102D** (to both, in the brake system **100** shown in the Figures). This manual push-through may be used in a backup or “failed” situation, when, for some reason, the power transmission unit **104** is unable to provide fluid to those selected wheel brakes. The manual push-through may also or instead be used to test system capacity as a part of normal operation, though such use case will not be discussed further herein.

(29) In order to facilitate operation of the brake system **100** in the manual push-through mode, a normally open base brake iso valve **152** is provided. The base brake iso valve **152** is energized

closed during normal non-failure braking mode. The base brake iso valve **152** can then be opened, either electrically or by cessation of electrical power, to place the master cylinder **110** into fluid communication with at least one of the fluid separators **142B**, **142D** during manual push-through braking mode. The fluid separators **142B**, **142D** can then be used to help switch the front wheel brakes **102B**, **102D** into their manual push-through mode of operation, as described above. The base brake iso valve **152** may include a check valve feature, as indicated schematically in the Figures, which seals only in one direction across the valve **152**.

(30) At least one electronic control unit (“ECU”) **154** may be provided for selectively controlling at least one of the power transmission unit **104**, one or more of the plurality of iso/dump valve arrangements **134**, and the base brake iso valve **152** responsive to the brake signal generated by the master cylinder **110** responsive to pressure upon the brake pedal **114** by the driver of the vehicle. As shown in the Figures, the brake system **100** may have a first electronic control unit **154A**, as well as a second electronic control unit **154B** for selectively controlling at least one of the power transmission unit **104**, one or more of the plurality of iso/dump valve arrangements **134**, and the base brake iso valve **152** responsive to the brake signal. The ECUs **154A**, **154B** may each control different ones of the electrically powered and/or controlled portions of the brake system **100**, or may each be configured to control one or more of the same component(s) of the brake system **100**, at different times, such as by providing redundant control in the event of failure of one of the ECUs **154A**, **154B**. The iso/dump valve arrangements **134** may each be controlled, for example by the first and/or second ECU **154A**, **154B**.

(31) The ECUs **154A**, **154B** may include microprocessors and other electrical circuitry. The ECUs **154A**, **154B** receive various signals, process signals, and control the operation of various electrical components of the brake system **100** in response to the received signals, in a wired and/or wireless manner. The ECUs **154A**, **154B** can be connected to various sensors such as the reservoir fluid level sensor **108**, pressure sensors, travel sensors, switches, wheel speed sensors, and steering angle sensors. The ECUs **154A**, **154B** may also be connected to an external module (not shown) for receiving information related to yaw rate, lateral acceleration, longitudinal acceleration of the vehicle, or other characteristics of vehicle operation for any reason, such as, but not limited to, controlling the brake system **100** during vehicle braking, stability operation, or other modes of operation. Additionally, the ECUs **154A**, **154B** may be connected to the instrument cluster for collecting and supplying information related to warning indicators such as an ABS warning light, a brake fluid level warning light, and a traction control/vehicle stability control indicator light.

(32) As shown (optionally, as indicated in dashed lines) in the Figures, a pair of rear brake motors **156A**, **156C** may be provided for selectively electrically actuating respective rear wheel brakes **102A** and **102C**, at least in a parking brake mode which acts in a complementary manner to the hydraulically actuated surface brakes **102A**, **102C**. Whether or not the rear brake motors **156A**, **156C** function during normal non-failure braking as parking brakes, however, the rear brake motors **156A**, **156C** could be used in a backup mode as electrically actuated substitutes for the hydraulically operated pair of rear wheel brakes **102A**, **102C** when hydraulic operation of the rear wheel brakes is not available, for some reason. Accordingly, at least a selected electronic control unit **154A**, **154B** may be provided for controlling at least one of the power transmission unit **104** and the pair of rear brake motors **156** (e.g., in a normal non-failure braking mode), and the other electronic control unit **154A**, **154B** may be provided for controlling at least one of the base brake iso valve **152** and the pair of rear brake motors **156** (e.g., in a backup braking mode), though it is contemplated that a selected one of the electronic control units **154** could control the rear brake motors **156** in both modes of operation.

(33) With reference now to FIG. 2, the brake system **100** of FIG. 1 is shown as including separation of the various components (hydraulically if not also mechanically) into separate modules or units, with each module being schematically delineated by one of dashed-line boxes A and B. The delineation noted in FIG. 2 by boxes A and B indicates a physical separation of the components

thus indicated (i.e., by being located remote from one another within the brake system **100**, and at least apart from a common housing or block), which may be desirable in certain use environments. For example, the components in box A could be located immediately adjacent to a dashboard of the vehicle, with the linkage arm **122** extending through the vehicle firewall (not shown). The components in box A would then be hydraulically linked to “remotely” located components in box B. stated otherwise, and for example, the reservoir **106** and master cylinder **110** may be co-located and both spaced apart from the power transmission unit **104**, fluid separators **142**, and iso/dump valve arrangements **134**. As such, the brake system **100** shown in FIG. 2 may be suited for use in a particular vehicle configuration in which such physical separation of the box A module components from the box B module components is desirable.

(34) FIGS. 3-4 illustrate a second embodiment of a brake system **100**. The brake system **100** of FIGS. 3-4 is similar to the brake system **100** of FIGS. 1-2 and therefore, structures of FIGS. 3-4 that are the same as or similar to those described with reference to FIGS. 1-2 have the same reference numbers. Description of common elements and operation similar to those in the previously described first embodiment will not be repeated with respect to the second embodiment, but should instead be considered to be incorporated below by reference as appropriate.

(35) As shown in FIG. 3, the brake system **100** may include a secondary brake module **158** of any desired type interposed hydraulically between the master cylinder **110** and the base brake valve **152**. The secondary brake module **158** shown in FIG. 3 includes an electric motor **160** and at least one pump chamber of any desired type (shown here as at least one pump piston **162**) for cooperatively increasing fluid pressure between the master cylinder **110** and the base brake valve **152** when the brake system **100** is in at least one of normal non-failure braking mode and manual push-through mode. A secondary iso valve **164** may be provided to assist with directing hydraulic fluid among the pump pistons **162** and other components of the brake system **100**, as desired. It is contemplated that the secondary brake module **158** could also or instead include at least one of a single-acting plunger type power transmission unit, a dual-acting plunger type power transmission unit, or any other desired mechanism(s) or component(s) for selectively increasing fluid pressure. One of ordinary skill in the art can readily provide a suitable secondary brake module **158** for a particular use environment of the brake system **100**.

(36) The secondary brake module **158** may be configured for selectively providing pressurized hydraulic fluid for actuating the pair of front wheel brakes **1028** and **102D**, when the brake system **100** is in any mode in which a pressure “boost” to one or more corresponding wheel brake(s) **102** (as shown in the brake system **100** of FIG. 3, just the front wheel brakes **1028**, **102D**) is desirable. That is, the secondary brake module **158** could be used only when the brake system **100** is in the manual push-through mode, to supplement the pressure applied by the driver, via the brake pedal **114** of the master cylinder **110**. It is also contemplated, though, that the secondary brake module **158** could be used when the brake system **100** is in the normal non-failure braking mode, to provide a “backup boost” pressure build, supplemental to pressure achieved by the power transmission unit **104**. The secondary brake module **158** could also or instead act to translate a brake signal from an autonomous braking system source (not shown) into brake-application pressure directed to one or more wheel brakes **102**, even without a driver's foot pressure upon the brake pedal **114**.

(37) One of ordinary skill in the art will be able to assign all components of the brake system **100** to an appropriate one (or more) of the electronic control units **154A**, **154B** (when more than one are provided) for a particular brake system **100**, in order to achieve desired redundant and backstopping functionability. For example, it may be desirable for the secondary brake module **158** to be powered and/or controlled by a different electronic control unit **154** than that which powers and/or controls at least one of the power transmission unit **104**, iso/dump valve arrangements **134**, and/or base brake iso valve **152**.

(38) As shown by dashed-line boxes A and B in FIG. 3, the secondary brake module **158** may be

spaced physically apart from one or more of the power transmission unit **104**, master cylinder **110**, fluid separators **142**, and iso/dump valve arrangements **134**. This physical separation may be provided for any desirable reason, and appropriate hydraulic and/or electrical lines may be used to connect the various spaced-apart components, for operation of the brake system **100** as desired.

(39) With reference now to FIG. **4**, the brake system **100** of FIG. **3** is shown as including a differently arranged (from FIG. **3**) separation of the various components—hydraulically if not also mechanically—into separate modules or units. Each module is schematically delineated by one of dashed-line boxes A, B, and C. The delineation noted by boxes A and B in FIG. **3**—boxes A, B, and C in FIG. **4**—indicates a physical separation of the components thus indicated (i.e., by being located remote from one another within the brake system **100**, and at least apart from a common housing or block), which may be desirable in certain use environments. For example, the components in box A could be located immediately adjacent to a dashboard of the vehicle, with the linkage arm **122** extending through the vehicle firewall (not shown). The components in box A would then be hydraulically linked to “remotely” located components in box B and/or box C. Stated otherwise, and in the example shown in FIG. **4**, the reservoir **106** and master cylinder **110** may be co-located and both spaced apart from the power transmission unit **104**, fluid separators **142**, and iso/dump valve arrangements **134**, which may, in turn, also be spaced physically apart from the secondary brake module **158**. As such, the brake system **100** shown in FIG. **4** may be suited for use in a particular vehicle configuration in which such physical separation of the box A module components from both of the box B module components and box C module components is desirable.

(40) At least one filter **172** may be provided in any desired position(s) within the brake systems **100** of FIGS. **1-4**.

(41) Pressure of hydraulic fluid within the brake systems **100** of FIGS. **1-2** could be sensed, directly or indirectly, with a pressure sensor (shown generally as “P”) at any desired location(s) in the brake systems **100**. It is contemplated that the pressure sensor(s) will be in electrical communication with at least one of the primary and secondary ECUs **154A**, **154B**, in a wired or wireless manner, to communicate pressure signals thereto responsive to sensed fluid pressure(s).

(42) As used herein, the singular forms “a”, “an”, and “the” can include the plural forms as well, unless the context clearly indicates otherwise. It will be further understood that the terms “comprises” and/or “comprising”, as used herein, can specify the presence of stated features, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, steps, operations, elements, components, and/or groups thereof.

(43) As used herein, the term “and/or” can include any and all combinations of one or more of the associated listed items.

(44) It will be understood that when an element is referred to as being “on”, “attached” to, “connected” to, “coupled” with, “contacting”, “adjacent”, etc., another element, it can be directly on, attached to, connected to, coupled with, contacting, or adjacent the other element, or intervening elements may also be present. In contrast, when an element is referred to as being, for example, “directly on”, “directly attached” to, “directly connected” to, “directly coupled” with, “directly contacting”, or “directly adjacent” another element, there are no intervening elements present. It will also be appreciated by those of ordinary skill in the art that references to a structure or feature that is disposed “directly adjacent” another feature may have portions that overlap or underlie the adjacent feature, whereas a structure or feature that is disposed “adjacent” another feature might not have portions that overlap or underlie the adjacent feature.

(45) Spatially relative terms, such as “under”, “below”, “lower”, “over”, “upper”, “proximal”, “distal”, and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms can encompass different orientations of a device in use or operation, in addition to the orientation depicted in the figures. For example, if a device in the

figures is inverted, elements described as “under” or “beneath” other elements or features would then be oriented “over” the other elements or features.

(46) As used herein, the phrase “at least one of X and Y” can be interpreted to include X, Y, or a combination of X and Y. For example, if an element is described as having at least one of X and Y, the element may, at a particular time, include X, Y, or a combination of X and Y, the selection of which could vary from time to time. In contrast, the phrase “at least one of X” can be interpreted to include one or more Xs.

(47) It will be understood that, although the terms “first”, “second”, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. Thus, a “first” element discussed below could also be termed a “second” element without departing from the teachings of the present disclosure. The sequence of operations (or steps) is not limited to the order presented in the claims or figures unless specifically indicated otherwise.

(48) While aspects of this disclosure have been particularly shown and described with reference to the example aspects above, it will be understood by those of ordinary skill in the art that various additional aspects may be contemplated. For example, the specific methods described above for using the apparatus are merely illustrative; one of ordinary skill in the art could readily determine any number of tools, sequences of steps, or other means/options for placing the above-described apparatus, or components thereof, into positions substantively similar to those shown and described herein. In an effort to maintain clarity in the Figures, certain ones of duplicative components shown have not been specifically numbered, but one of ordinary skill in the art will realize, based upon the components that were numbered, the element numbers which should be associated with the unnumbered components; no differentiation between similar components is intended or implied solely by the presence or absence of an element number in the Figures. Any of the described structures and components could be integrally formed as a single unitary or monolithic piece or made up of separate sub-components, with either of these formations involving any suitable stock or bespoke components and/or any suitable material or combinations of materials. Any of the described structures and components could be disposable or reusable as desired for a particular use environment. Any component could be provided with a user-perceptible marking to indicate a material, configuration, at least one dimension, or the like pertaining to that component, the user-perceptible marking potentially aiding a user in selecting one component from an array of similar components for a particular use environment. A “predetermined” status may be determined at any time before the structures being manipulated actually reach that status, the “predetermination” being made as late as immediately before the structure achieves the predetermined status. The term “substantially” is used herein to indicate a quality that is largely, but not necessarily wholly, that which is specified—a “substantial” quality admits of the potential for some relatively minor inclusion of a non-quality item. Though certain components described herein are shown as having specific geometric shapes, all structures of this disclosure may have any suitable shapes, sizes, configurations, relative relationships, cross-sectional areas, or any other physical characteristics as desirable for a particular application. Any structures or features described with reference to one aspect or configuration could be provided, singly or in combination with other structures or features, to any other aspect or configuration, as it would be impractical to describe each of the aspects and configurations discussed herein as having all of the options discussed with respect to all of the other aspects and configurations. A device or method incorporating any of these features should be understood to fall under the scope of this disclosure as determined based upon the claims below and any equivalents thereof.

(49) Other aspects, objects, and advantages can be obtained from a study of the drawings, the disclosure, and the appended claims.

Claims

1. A brake system for actuating at least one of a pair of front wheel brakes and a pair of rear wheel brakes in normal non-failure and manual push-through modes, the system comprising: a reservoir; a master cylinder operable to provide a brake signal responsive to actuation of a brake pedal connected thereto, the master cylinder being selectively operable during a manual push-through mode by actuation of the brake pedal to generate brake actuating pressure to at least one output for hydraulically actuating the pair of front wheel brakes; a power transmission unit configured for selectively providing pressurized hydraulic fluid for actuating at least one of the pair of front wheel brakes and the pair of rear wheel brakes in a normal non-failure braking mode; a plurality of iso/dump valve arrangements, each iso/dump valve arrangement being hydraulically interposed between the power transmission unit and a corresponding one of the wheel brakes; a plurality of fluid separators, each fluid separator being hydraulically interposed between a selected one of the pair of front wheel brakes and a corresponding iso/dump valve arrangement, each fluid separator isolating the fluid within the corresponding iso/dump valve arrangement from the master cylinder; a normally open base brake iso valve, the base brake iso valve being energized closed during normal non-failure braking mode and the base brake iso valve being open to place the master cylinder into fluid communication with at least one of the fluid separators during manual push-through braking mode; and an electronic control unit for selectively controlling at least one of the power transmission unit, at least one of the plurality of iso/dump valve arrangements, and the base brake iso valve responsive to the brake signal.
2. The brake system of claim 1, wherein the primary power transmission unit is of a dual acting plunger type.
3. The brake system of claim 1, wherein the master cylinder includes a housing defining a longitudinally extending bore, and an MC spring within the bore.
4. The brake system of claim 1, wherein the master cylinder includes an integrated pedal simulator for providing predetermined brake pedal response.
5. The brake system of claim 4, wherein the pedal simulator is coaxial with a longitudinally extending bore of the master cylinder.
6. The brake system of claim 1, including a normally closed DAP valve interposed hydraulically between the power transmission unit and the plurality of iso/dump valve arrangements.
7. The brake system of claim 1, wherein at least one iso/dump valve arrangement includes a normally open dump valve for each wheel brake of the pair of rear wheel brakes.
8. The brake system of claim 1, wherein at least one iso/dump valve arrangement includes a normally closed dump valve for each wheel brake of the pair of front wheel brakes.
9. The brake system of claim 1, wherein each fluid separator includes a fluid separator piston spring-biased toward a first end of a fluid separator chamber.
10. The brake system of claim 9, wherein each fluid separator piston travels longitudinally between a first end and a second end of each fluid separator chamber in normal non-failure braking mode, and wherein each fluid separator piston is maintained substantially adjacent each first end in manual push-through braking mode.
11. The brake system of claim 1, wherein the electronic control unit is a first electronic control unit, and wherein the brake system includes a second electronic control unit for selectively controlling at least one of the power transmission unit, at least one of the plurality of iso/dump valve arrangements, and the base brake iso valve responsive to the brake signal.
12. The brake system of claim 1, wherein the base brake iso valve includes a check valve feature sealing only in one direction.
13. The brake system of claim 1, including a secondary brake module interposed hydraulically between the master cylinder and the base brake valve.

14. The brake system of claim 13, wherein the secondary brake module includes an electric motor and at least one pump chamber for cooperatively increasing fluid pressure between the master cylinder and the base brake valve during at least one of normal non-failure braking mode and manual push-through mode.
 15. The brake system of claim 1, wherein the reservoir and master cylinder are co-located and both spaced apart from the power transmission unit, fluid separators, and iso/dump valve arrangements.
 16. The brake system of claim 13, wherein the secondary brake module is spaced apart from the power transmission unit, master cylinder, fluid separators, and iso/dump valve arrangements.
 17. The brake system of claim 16, wherein the reservoir and master cylinder are co-located and both spaced apart from the power transmission unit and iso/dump valve arrangements.
 18. The brake system of claim 1, wherein the base brake iso valve is in fluid communication with the master cylinder and is fluidically spaced from the power transmission unit.
 19. The brake system of claim 1, wherein each fluid separator is in fluid communication with the corresponding iso/dump valve arrangement through a first conduit and in fluid communication with the base brake iso valve and the master cylinder through a second conduit hydraulically spaced from the first conduit.
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